



UNIVERSIDADE DE COIMBRA
FACULDADE DE CIÊNCIAS E TECNOLOGIA
Departamento de Engenharia Informática

Advanced Data Visualization (MECD)
Regular Season

08/06/2026

NAME: _____

Question Group 1: (20%)

A) Explain the main difference between traditional visualization and visual analytics.

B) Comment the following affirmation: “Visualization of Information and Geographic Visualization use the same techniques and methods”.

C) Comment the following affirmation: “Visualization helps humans go beyond their limits”.

D) Comment the following affirmation: “Analysis always needs visualization for data analysis”.

Question Group 2: (20%)

A) For each variable represented in the visualization (Figure 1) identify the following:
(i) what **data attribute** is represented; (ii) its **data type**; (iii) what visual **mark** is used;
(iv) and **channel** (visual property) is mapped to.

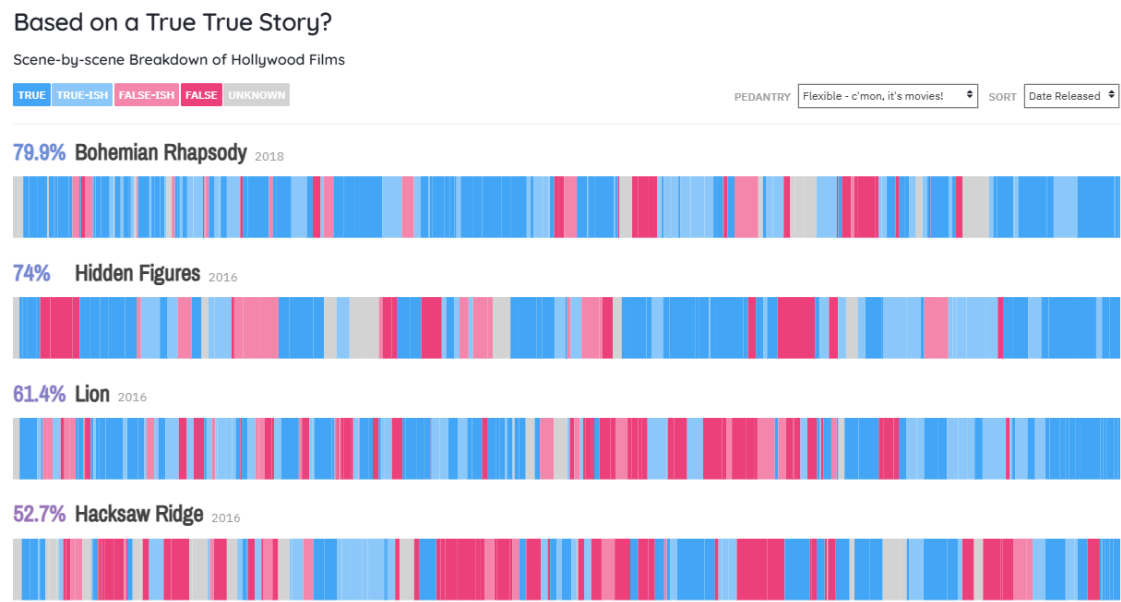


Figura 1

<https://informationisbeautiful.net/visualizations/based-on-a-true-true-story/>

Data Variable	Data Type	Mark	Channel

Question Group 3: (15 %)

A) Geovisualization: Describe Area Cartogram Technique.

A large, empty rectangular box with a black border, intended for the student to write their answer to question A.

B) Time series: Describe Horizon Graph technique. Sketch example.

A large, empty rectangular box with a black border, intended for the student to write their answer to question B, including a sketch.

C) Tabular data: Describe Parallel Coordinates Technique.

A large, empty rectangular box with a black border, intended for the student to write their answer to question C.

Question Group 4: (20%)

Considering the visualization in **Figure 2**, provide a critical analysis, listing all the identified problems.

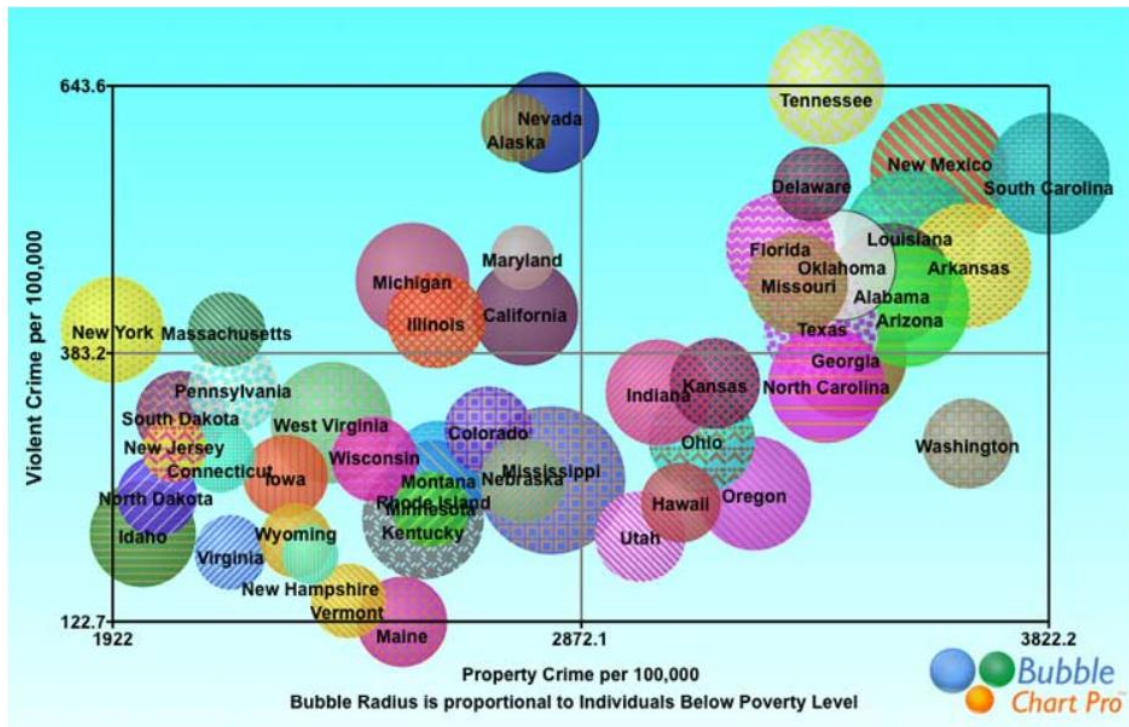


Figure 2

A) Provide a critical analysis.

B) Propose and sketch na alternative of visualization.



Question group 5: (25%)

EcoTracker, an environmental monitoring company, requires an analytical visualization tool to track animal species in real-time and identify zones with "conservation risks" (e.g., multiple species abruptly abandoning an area, or anomalous overcrowding in shrinking habitats).

Your visualization solution must allow analysts to answer the following:

- a) Analyze the movement patterns of different species over time.
- b) Identify spatio-temporal patterns of habitat usage.
- c) Identify areas with high density of animal activity, focusing on specific species, number of individuals, and highlighting the affected geographic zones.

Data: You may assume you have access to **any datasets or variables** you deem necessary to solve this problem.

Task: Propose an **innovative analytical dashboard** that effectively addresses these three goals. Sketch and describe your layout, and clearly justify your design choices.

